

Linear Regression from Scratch: Inference vs. Prediction

Kavi Mookherjee Amodt

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Abstract

Inferential vs predictive notions of the same statistical process: linear regression.

- 1 The two cultures: inference vs prediction**
- 2 Linear regression is just projecting onto $\text{span}(\mathbf{X})$**
- 3 Gradient descent for OLS**
- 4 OLS is MLE under Gaussian noise**
- 5 Logistic regression doesn't have closed form**
- 6 Regularizing is biasing on purpose**